1) for loop:

If we want to execute some action for every element present in some sequence(it may be string or collection)then we should go for for loop.

Syntax:

```
for x in sequence
  :
     body
```

where sequence can be string or any collection.
Body will be executed for every element present

in the sequence.

Q1. PRINT 1 TO 10 NOS USING FOR LOOP

Q2. PRINT 1 TO N NOS USING FOR LOOP

Q3. PRINT ODD NOS B/W 1 TO 10 USING FOR LOOP

Q4. PRINT EVEN NOS B/W 1 TO 10 USING FOR LOOP

Q5. PRINT N TO 1 NOS USING FOR LOOP

SUM OF N NOS

n = 4

1 + 2 + 3 + 4 = 10

s = s + i ; s = 0 ;

Q6. WAP FOR SUM OF N NOS

Q7. WAP FOR FACTORIAL

trace:- f = 1 , n = 4

for i = 1 to 4

f = f * i

i = 1 f = 1 * 1 = 1

i = 2 f = 1 * 2 = 2

i = 3 f = 2 * 3 = 6

i = 4 f = 6 * 4 = 24

fact = 24

----

n = 6 f = 720

n = 7 f = 5040

Q8. MODIFIED FACTORIAL Not accept negative value

Q9. MULTIPLICATION TABLE

Q10 FIBONACCI SERIES

0 1 1 2 3 5 8 13 .....

given

e.g. n = 4 0 1 1 2 3

0th 1st 2nd 3rd 4th

```
a, b = 0, 1
print("   ",a,"   ",b,end="")
for i in range(1, n):
    c = a + b
    print("   ",c, end="")
    a = b # imp ( new a and b )
    b = c
```

Q11

CONTINUE STATEMENT

WAP FOR PRINT 1 TO 100 NO. EXCEPT 20 TO 30.

Q12.

Example OF break STATEMENT

PRIME NO. [else with for loop]

```
n = int(input("ENTER NO\n"))

for i in range(2, n):
```

```
    if n % i == 0:
        print("NOT PRIME NO OR COMPOSITE NO.")
        break
else:
    print("PRIME NO")
```

---

trace:- 1.

n = 5 ;

for i = 2 to 4

i = 2 t = 5 % 2 = 1

i = 3 t = 5 % 3 = 2

i = 4 t = 5 % 4 = 1

PRIME NO.

---

2. n = 9

for i = 2 to 8

i = 2 t = 9 % 2 = 1

i = 3 t = 9 % 3 = 0

break ;

i = 4,5,6,7,8 X

NOT PRIME NO. OR COMPOSITE NO.

---

3. n = 2 , t = 1

IMP for i = 2 to 1 X

Q13

LCM (Least Common Multiple)

LCM of two numbers is the smallest number which can be divided by both numbers.

For example, LCM of 15 and 20 is 60,

and LCM of 5 and 7 is 35.

WAP TO FIND LCM

```
print("Enter two positive integers")
n1, n2 = map(int, input().split())

# maximum number between n1 and n2 is stored in max
max_num = max(n1, n2)

while True:  # infinite loop
   if max_num % n1 == 0 and max_num % n2 == 0:
      print(f"The LCM of {n1} and {n2} is {max_num}.")
      break
   max_num += 1
```

trace :

n1 = 2 , n2 = 3

max = max(n1, n2)

max = 3	**max % n1 == 0**	**and**	**max % n2 == 0**
max = 3	**3 % 2 == 0**	**and**	**3 % 3 == 0 F**
max = 4	**4 % 2 == 0**	**and**	**4 % 3 == 0 F**
max = 5	**5 % 2 == 0**	**and**	**5 % 3 == 0 F**
max = 6	**6 % 2 == 0**	**and**	**6 % 3 == 0 T**

break LCM = max = 6

Highest Common Factor (HCF) : OR

Greatest Common Divisor (GCD) :

The HCF or GCD of two integers is the largest integer

that can exactly divide both numbers (without a remainder).

For example, HCF of 12 and 18 is 6. AND

HCF of 16 and 32 is 16. AND

HCF of 12 and 15 is 3.

Q14. WAP to Find HCF OR GCD of two Numbers

HCF of 12 and 18 is 6. n1 = 12 , n2 = 18

n1 % i ==0	**and**	**n2 % i == 0**	**gcd = i**
i = 1 12 % 1 == 0	**and**	**18 % 1 == 0 T**	**gcd = 1**
i = 2 12 % 2 == 0	**and**	**18 % 2 == 0 T**	**gcd = 2**
i = 3 12 % 3 == 0	**and**	**18 % 3 == 0 T**	**gcd = 3**
i = 4 12 % 4 == 0	**and**	**18 % 4 == 0 F**	
i = 5 12 % 5 == 0	**and**	**18 % 5 == 0 F**	
i = 6 12 % 6 == 0	**and**	**18 % 6 == 0 T**	**gcd = 6**

. . .

i = 12 12 % 12== 0 && 18 % 12 == 0 F

ANS :- GCD = 6

----------END----------